

SaaS Customer Churn & Revenue Analysis

SQL Portfolio Project • RavenStack SaaS Dataset

Executive Summary

This project analyzes a SaaS customer base using SQL to understand customer segmentation, subscription revenue, churn behavior, support activity, and revenue associated with churned customers. The analysis uses five RavenStack tables covering accounts, subscriptions, feature usage, support tickets, and churn events.

Metric	Value
Customers	500
Subscription records	5,000
Churn events	600
Feature usage records	25,000
Support tickets	2,000
Countries	—

Key Findings

- Pro is the largest customer plan with 178 customers; Basic has 168 and Enterprise 154.
- Enterprise contributes the largest revenue: total MRR of 8,473,221 and total ARR of 101,678,652.
- Enterprise also has the highest average subscription MRR at 4,917.71.
- Pro has the highest plan-level churn rate at 72.47%, followed by Enterprise at 70.13% and Basic at 68.45%.
- DevTools has the highest industry churn rate at 73.45%; HealthTech is lowest at 66.67%.
- Churned customers average 6.79 support tickets each versus 4.09 for active customers—about 66% higher.
- Subscription revenue associated with churned customers totals approximately 13,191,658 MRR and 158,299,896 ARR in the analysis output.

Customer & Revenue Analysis

Plan	Customers	Total MRR	Total ARR	Avg MRR
Enterprise	154	8,473,221	101,678,652	4,917.71
Pro	178	2,105,089	25,261,068	1,256.77
Basic	168	760,437	9,125,244	474.68

Industry customer distribution: DevTools 113, FinTech 112, Cybersecurity 100, HealthTech 96, and EdTech 79.

Trial status: 97 customers are trials and 403 are non-trials, meaning trials represent 19.4% of the customer base.

Churn Analysis

Plan	Churned	Customers	Churn Rate
Pro	129	178	72.47%
Enterprise	108	154	70.13%

Basic	115	168	68.45%
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Industry churn rates: DevTools 73.45%, EdTech 72.15%, Cybersecurity 72.00%, FinTech 67.86%, HealthTech 66.67%.

The dataset reports a high overall churn level. Before treating these percentages as a real-world benchmark, the churn-event definition and observation period should be validated.

Support Activity & Churn

Active customers: 148 customers and 605 support tickets, averaging 4.09 tickets per customer.

Churned customers: 352 customers and 2,389 support tickets, averaging 6.79 tickets per customer.

Interpretation: churned customers show substantially higher support activity. This is an association in the dataset, not proof that support tickets cause churn.

Revenue Associated with Churn

The final SQL analysis returned 13,191,658 MRR and 158,299,896 ARR when subscription records were joined to churn events by account_id.

Because the dataset contains multiple subscription records per customer, this should be described as revenue associated with churned customers/subscription records rather than guaranteed lost revenue.

Business Recommendations

1. Prioritize retention efforts for DevTools customers because the segment has the highest industry churn rate.
2. Investigate Pro-plan churn because Pro has the highest plan-level churn rate.
3. Use elevated support activity as an early-warning signal and review recurring issues for high-ticket customers.
4. Protect high-value Enterprise accounts because Enterprise has substantially higher MRR and ARR per subscription.
5. Combine feature usage, support activity, subscription history, and churn events in a future predictive churn model.

SQL Techniques Demonstrated

COUNT / COUNT DISTINCT • SUM • AVG • GROUP BY • ORDER BY • CASE WHEN • INNER JOIN • LEFT JOIN • ROUND • customer segmentation • churn-rate calculations • revenue analysis.

Project Conclusion

The analysis demonstrates how SQL can transform operational SaaS data into actionable business insights. The strongest signals are high churn across several segments, elevated churn in DevTools and Pro customers, higher support activity among churned customers, and substantial subscription revenue associated with churned accounts. These findings provide a clear foundation for retention strategy and a future Power BI dashboard or predictive churn model.